RESEARCH PAPER

Exploratory Data Analysis and Predictive Forecasting of Care Load & Placement Demand

Insights, Decision-Support Implications, and Recommendations

Paper details	Description
Research type	Applied analytics / forecasting prototype
Study focus	Daily care-load patterns, intake–exit imbalance, discharge demand
Analytical approach	EDA, time-series visualization, baseline forecasting, holdout evaluation
Data status	Simulated demonstration data; no official operational records analyzed

DATA INTEGRITY STATEMENT

This paper describes the analytical design and interprets the prototype outputs conceptually. Since the dashboard currently uses simulated data, it does not report empirical HHS findings, official statistics, or validated operational forecast performance.

Abstract

This paper presents an exploratory data analysis (EDA) and predictive forecasting framework for understanding care load and placement demand. The project focuses on three primary analytical questions: how the number of children in HHS care may evolve, whether intake and exits may become imbalanced, and what short-term discharge demand may look like. Secondary aims include early warning, uncertainty communication, and comparison of forecasting approaches. The prototype uses a synthetic daily time series and an interactive Streamlit dashboard to demonstrate data summaries, trend plots, rolling averages, forecast trajectories, approximate residual-based ranges, capacity-threshold scenarios, and chronological holdout metrics. The EDA framework examines temporal coverage, distributions, missingness, outliers, trend, variability, and relationships among flow and stock measures. Because the data are simulated and not reconciled to official operational records, the insights are methodological and illustrative rather than empirical. The paper concludes with recommendations for data governance, flow reconciliation, repeated time-series validation, calibrated uncertainty, and responsible operational adoption.

Keywords: exploratory data analysis; care load; placement demand; time-series forecasting; discharge demand; capacity planning; uncertainty; decision support

1. Introduction

Care and placement systems operate in a dynamic environment in which the number of children requiring care can change as new cases enter, transfers occur, and children exit through discharge or other pathways. A current count is useful for situational awareness, but it does not by itself indicate the likely workload over the coming days. Analytical methods that describe historical patterns and estimate near-term changes can supplement planning and review.

This study develops a prototype analytical workflow titled “Predictive Forecasting of Care Load & Placement Demand.” It combines EDA with a short-horizon forecasting dashboard. The emphasis is not on asserting a definitive operational forecast, but on establishing a transparent workflow through which trends, forecast errors, and potential capacity pressure can be examined.

2. Research Problem and Questions

The central problem is the limited forward visibility that may arise when care-load information is reviewed only as a series of current or historical counts. The project asks:

1. What temporal patterns and variability are visible in care-load and discharge measures?
2. How can intake/inflow and exits be compared to characterize a potential flow imbalance?
3. Can a short-term forecast provide a useful planning signal for care load and discharge demand?
4. How should forecast uncertainty and potential threshold crossings be communicated?
5. How can simple statistical baselines be compared fairly with more advanced forecasting models?

3. Objectives

3.1 Primary objectives

- Forecast the number of children in HHS care.
- Estimate future imbalance between intake and exits.
- Predict short-term discharge demand.

3.2 Secondary objectives

- Provide early warnings for healthcare planners.
- Quantify forecast uncertainty.
- Compare statistical versus machine-learning forecasting approaches.

4. Data and Study Design

The current dashboard operates without a CSV upload by generating synthetic daily observations. Its conceptual measures include date, children apprehended and placed in CBP custody, children in CBP custody, children transferred out of CBP custody, children in HHS care, children discharged from HHS care, and a simulated HHS inflow series. The synthetic series are intended to support demonstration of charts and methods; they are not official records.

Data element	Analytical role	Interpretive caution
Children in HHS care	Stock measure; primary care-load target	Stock definition and reference time must be consistent.
Children discharged from HHS care	Flow measure; discharge-demand target	Daily event counts may be volatile and may include calendar effects.
Apprehensions / CBP custody	Intake and custody context	Operational definitions and timing require documentation.
Transfers out of CBP custody	Flow measure	A transfer is not necessarily an HHS-care exit; confirm pathway definitions.
Simulated HHS inflow	Illustrative flow-balance input	Generated independently; not reconciled to care-load stock.

A real study should specify the observation unit (daily), time zone and cut-off time, reporting lag, treatment of revised records, and whether counts represent events during a day or stock at a point in time. Data access should be authorized, and any personal identifiers should be excluded from the analytical dataset unless strictly necessary and governed.

5. Methodology

5.1 Exploratory Data Analysis framework

EDA dimension	Recommended analysis	Purpose
Structure and coverage	Row count, date span, frequency, duplicate dates	Confirm a usable daily time index.
Completeness	Missing dates, null counts, reporting gaps	Identify gaps that may bias trends or forecasts.
Validity	Negative counts, non-integer values, impossible jumps	Flag values requiring source verification.
Distribution	Mean, median, standard deviation, quantiles, skewness	Understand typical levels and variability.
Outliers	IQR flags, robust z-scores, time-series inspection	Separate true surges from data errors.
Trend and seasonality	Daily plots, rolling mean, day-of-week/month patterns	Identify direction and recurring temporal structure.
Relationships	Correlation and lagged correlation among measures	Explore association while avoiding causal claims.
Flow reconciliation	Opening stock + inflows − exits ± adjustments	Test consistency between stock and flow definitions.

5.2 Trend and smoothing

The dashboard displays daily observations and a configurable rolling mean. Smoothing helps reveal the underlying direction when daily values fluctuate. The smoothing window should be selected to match the operational planning horizon and should not conceal abrupt but meaningful changes.

5.3 Forecast construction

The prototype uses linear regression over the time index and blends the projected trend with a recent 14-day mean. Negative predictions are truncated at zero because the targets are counts. This blend is a transparent demonstration baseline; it may underrepresent nonlinear patterns, abrupt shifts, and seasonal behavior.

5.4 Validation and metrics

Forecast evaluation should preserve chronology. The prototype uses a later holdout segment to evaluate predictions trained on earlier observations. MAE measures average absolute error in count units, RMSE penalizes larger errors more strongly, and MAPE expresses relative error but is undefined or unstable when actual values are zero or near zero. The dashboard's accuracy percentage, defined as 100 minus MAPE and bounded to 0–100%, is a project proxy rather than a universally standardized accuracy metric.

5.5 Uncertainty and capacity scenarios

The prototype displays an approximate residual-based forecast range. Such a range communicates uncertainty, but it is not a calibrated prediction interval unless coverage is tested on independent data. A capacity threshold is a scenario input. The proportion of forecast days whose upper estimate exceeds the threshold is an exposure indicator, not a statistically calibrated probability of capacity breach.

6. EDA Findings and Insights

6.1 What the current prototype supports

Because the active dataset is synthetic and no empirical output table has been supplied for this paper, the following are analytical interpretations and questions the dashboard is designed to answer—not observed HHS findings.

- Care-load trajectory: the time-series plot and rolling mean allow users to determine whether the selected period appears to rise, decline, or remain broadly stable.
- Volatility: the distribution and variability summaries help distinguish a relatively steady series from one with large daily changes.
- Discharge behavior: daily discharge counts can be examined for recurring patterns, isolated spikes, and forecast error.
- Flow balance: comparing inflow and discharges highlights periods in which the flow difference is positive or negative; conclusions depend on consistent event definitions.
- Capacity scenario: a user-defined threshold supports a conditional review of projected load, but only a formally approved capacity value makes the result operationally meaningful.
- Forecast reliability: holdout MAE/RMSE and comparison charts provide evidence about relative baseline performance on the test period, not a guarantee of future performance.

6.2 EDA interpretation principles

A visible upward trend should not automatically be interpreted as a causal effect. Correlation between two measures may reflect shared time trends or overlapping definitions. An outlier may represent a genuine operational event rather than a data error. Missing days should not be silently treated as zero unless zero activity is confirmed. These principles are especially important when interpreting care and placement counts.

7. Insights for Decision Support

Analytical signal	Potential planning interpretation	Required verification
Sustained positive inflow-minus-exit balance	Care load may increase if the imbalance persists	Reconcile all entry, transfer, discharge and adjustment categories.
Increasing care-load forecast	Review near-term staffing and placement availability	Validate against latest operational counts and model error.
Forecast upper range nearing capacity	Consider a precautionary scenario review	Use approved capacity definition and calibrated interval coverage.
High discharge forecast	Prepare discharge coordination and downstream planning	Confirm case-level readiness; aggregate forecast does not determine individual outcomes.
Large holdout errors or unstable model ranking	Forecast may be insufficiently reliable for decisions	Run multiple rolling-origin tests and assess drift.

8. Recommendations

8.1 Data and governance

1. Replace simulated data with authorized, quality-checked historical observations before publishing operational findings.
2. Create a data dictionary that defines each measure, its unit, event timing, stock cut-off, and inclusion/exclusion rules.
3. Reconcile stock and flow measures and document transfers, adjustments, and pathways that do not map directly to care exits.
4. Add automated validation for duplicates, missing dates, nulls, negative values, impossible ranges, and reporting delays.

8.2 Analytical methods

1. Use rolling-origin or walk-forward validation over several forecast origins rather than relying on one holdout.
2. Compare seasonal naïve, moving-average, exponential-smoothing/ARIMA, and machine-learning models such as Random Forest or Gradient Boosting on identical chronological folds.
3. Report MAE and RMSE in count units, and use MAPE only when appropriate; consider WAPE or sMAPE for zero-heavy data.
4. Calibrate forecast intervals and report empirical coverage and interval width.
5. Test forecast performance by horizon and by periods of unusually high demand.

8.3 Operational implementation

1. Define surge thresholds and capacity limits with authorized program owners; avoid treating arbitrary scenario values as policy.
2. Present forecasts as decision-support signals, not automatic decisions about individual children or placements.
3. Establish a review cadence, forecast ownership, and an escalation protocol for threshold alerts.
4. Monitor model drift and maintain a record of model versions, data cut-off times, and forecast outputs.
5. Ensure access controls, auditability, privacy safeguards, and appropriate review before sensitive data are used.

9. Limitations

- No official HHS or operational historical dataset was analyzed for this paper; the dashboard uses simulated values.
- No empirical numerical EDA statistics, model scores, or validated insights can be claimed from the present source context.
- The synthetic inflow is independently generated and does not necessarily reconcile to care-load stock.
- The linear trend/recent-mean blend is simple and may fail under structural breaks or complex seasonality.
- The residual-based range and KPI proxies are not calibrated or standardized.
- The prototype's baseline comparison does not yet fulfill a complete statistical-versus-machine-learning benchmark.

10. Conclusion

This paper sets out an EDA and forecasting framework for care-load and placement-demand planning. The dashboard supports time-series exploration, discharge-demand forecasting, flow comparison, uncertainty visualization, and capacity-threshold scenarios. Its principal contribution is a transparent prototype workflow that connects analytical objectives to interpretable outputs. The current findings are methodological because the data are simulated. A defensible operational research study should use validated historical data, reconcile flow definitions, apply repeated chronological evaluation, calibrate uncertainty, and agree on KPI and capacity definitions with stakeholders.

Suggested Figure and Table Set for a Full Empirical Paper

- Figure 1. Daily HHS care-load series with rolling mean.
- Figure 2. Distribution of daily discharge counts (histogram/boxplot).
- Figure 3. Daily inflow and discharge comparison with net-flow line.
- Figure 4. Forecast versus actual values on chronological holdout.
- Figure 5. Forecast horizon with calibrated prediction interval and capacity threshold.
- Table 1. Data dictionary and data-quality summary.
- Table 2. Descriptive statistics by measure.
- Table 3. Outlier dates and source-verification outcomes.
- Table 4. Baseline and ML model comparison across rolling-origin folds.
- Table 5. KPI outcomes, definition, and decision threshold.

References / Methodological Sources

This draft does not cite external studies because no source list or external research materials were supplied. For a formal research submission, add references for time-series forecasting, rolling-origin validation, forecast accuracy metrics, prediction interval calibration, and the applicable program data definitions. Cite the exact source documents used for operational definitions and data.